

Agile Sprint Planning & User Story Workshop

Smart Airport Passenger Flow and Queue Management System

Student Name: _M.hanif riyaz

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1. Project Vision and Stakeholders

1.1 Project Vision

To build an AI-powered, real-time Smart Airport Passenger Flow and Queue Management System that integrates computer vision, IoT sensors, and cloud-based analytics to monitor passenger density across all airport zones, predict congestion before it occurs, optimize resource allocation dynamically, and deliver actionable alerts to airport operations teams – ultimately reducing passenger waiting times, minimizing flight delays, and elevating the overall travel experience.

1.2 Project Objectives

- Monitor passenger movement and density in real time across check-in halls, security checkpoints, immigration counters, boarding gates, and retail zones.
- Predict congestion hotspots using AI-based time-series and spatial analytics models.
- Optimize queue management by dynamically recommending lane openings, closures, and staffing adjustments.
- Allocate airport resources (staff, equipment, gates) based on predicted passenger loads.
- Generate real-time congestion alerts with severity levels and recommended actions.
- Analyze historical passenger flow patterns to inform strategic planning and infrastructure investments.
- Improve security checkpoint throughput while maintaining compliance with safety protocols.
- Enhance passenger satisfaction through reduced wait times and proactive communication.
- Provide executive dashboards for airport management with KPIs on flow efficiency, dwell times, and service levels.
- Ensure data privacy compliance for camera-based monitoring through anonymization and access controls.

1.3 Key Stakeholders

Stakeholder	Role in the Project
Airport Operations Manager	Primary decision-maker for resource allocation, congestion response, and operational policy based on system recommendations.
Terminal Supervisors	Monitor zone-specific dashboards and coordinate frontline staff deployment in response to alerts.
Security Checkpoint Managers	Use throughput analytics and queue predictions to optimize lane staffing and equipment utilization.
Ground Handling Staff	Receive mobile alerts for boarding gate congestion and passenger flow anomalies.
Airline Operations Coordinators	Consume boarding-gate flow data to adjust boarding sequences and minimize turnaround delays.
Immigration & Customs Officers	Leverage queue predictions to open or close immigration lanes proactively.
Retail & Concessions Managers	Analyze foot traffic patterns to optimize store staffing and promotional timing.
Passenger Experience Officers	Monitor passenger satisfaction metrics and dwell-time analytics to identify service gaps.
IT / Infrastructure Engineers	Manage camera networks, IoT sensor deployments, cloud infrastructure, and system uptime.
Data Scientists / AI Engineers	Build, train, validate, and monitor the congestion prediction and computer vision models.
System Administrators	Manage user access, security, platform maintenance, and integration health.
Privacy & Compliance Officer	Ensures camera-based monitoring meets data protection regulations (GDPR, local privacy laws).
Product Owner	Owns and prioritizes the product backlog based on stakeholder value and strategic alignment.
Scrum Master	Facilitates Agile ceremonies, tracks velocity, removes impediments, and coaches the team.
Airport Authority / Executive Leadership	Receives executive dashboards and uses insights for strategic planning and investment decisions.

1.4 Stakeholder RACI Matrix

The matrix below clarifies decision-making authority across the platform's core activities. R = Responsible, A = Accountable, C = Consulted, I = Informed.

Activity	Ops Manager	Terminal Supervisors	Security Mgrs	IT/Data Team	Compliance	Airlines
Real-Time Passenger Monitoring	A	R	R	R	I	I
AI Model Development & Validation	C	I	I	R/A	C	I
Queue Optimization Recommendations	A	R	R	R	I	C
Alert Generation & Response	A	R	R	R	I	I
Resource Allocation Decisions	A	R	R	C	I	C
Security Checkpoint Optimization	C	C	A/R	R	C	I
Data Privacy & Compliance	C	I	I	R	A	I
Executive Reporting & Analytics	A	C	C	R	C	I

1.5 Project Scope

In Scope

- Real-time ingestion of video feeds from cameras across terminal zones (check-in, security, immigration, gates, retail).
- Integration with IoT sensors (infrared counters, Bluetooth/Wi-Fi tracking beacons, weight sensors at queue barriers).
- AI-based passenger density estimation using computer vision (anonymized, privacy-preserving).
- Time-series congestion prediction models with 15-minute, 30-minute, and 1-

hour forecast horizons.

- Dynamic queue management recommendations (lane openings, staff redeployment).
- Multi-channel alerting (dashboard, mobile push, SMS) with configurable severity thresholds.
- Historical flow analytics and pattern visualization dashboards.
- Role-based access control and comprehensive audit logging.
- Executive KPI dashboards with service-level tracking.
- API integration with flight information display systems (FIDS) for arrival/departure context.

Out of Scope (Current Phase)

- Facial recognition or biometric identification (privacy constraint; system uses anonymized silhouette detection only).
- Direct integration with airline reservation systems (PNR data) — flight manifest data is consumed via FIDS only.
- Automated physical control of gates or barriers (the system recommends; humans or separate systems execute).
- Baggage handling flow optimization (separate system; future integration candidate).
- Retail point-of-sale integration (foot traffic only; no purchase data).

2. Epics and User Stories

The product requirements are organized into eight Epics, each representing a major functional area of the system. User stories follow the standard Agile format: "As a <user>, I want <feature>, so that <benefit>." Each story carries its MoSCoW priority, business value context, dependencies, and estimated story points.

EP-01: Real-Time Passenger Monitoring & Data Ingestion

Epic Goal: Capture and process real-time passenger density data from cameras and IoT sensors across all terminal zones.

Without continuous, accurate density data, every downstream capability — prediction, optimization, alerting — operates blind. This epic establishes the foundational data pipeline that feeds the entire platform, ensuring low-latency ingestion and reliable processing of high-volume video and sensor streams.

- US-01 (Must Have, 8 pts): As an IT infrastructure engineer, I want the system to ingest video feeds from terminal cameras in real time, so that passenger density can be continuously monitored without manual observation.
- US-02 (Must Have, 5 pts): As an IT infrastructure engineer, I want the system to receive data from IoT sensors (infrared counters, beacons) at queue entry and exit points, so that passenger counts can be validated and cross-

referenced with camera data.

- US-03 (Must Have, 8 pts): As a data scientist, I want a computer vision pipeline that estimates passenger density from video feeds using anonymized silhouette detection, so that privacy is preserved while providing accurate crowd counts.
- US-04 (Should Have, 5 pts): As an operations manager, I want zone-level passenger counts updated at least every 30 seconds, so that congestion changes are visible in near real time.
- US-05 (Should Have, 3 pts): As a system administrator, I want the ingestion pipeline to log data-quality issues (dropped frames, sensor timeouts), so that infrastructure problems can be diagnosed quickly.

EP-02: AI-Based Congestion Prediction

Epic Goal: Apply AI/ML models to current and historical flow data to forecast congestion before it materializes.

Reacting to congestion after it forms is too late — passengers are already frustrated and flights are already delayed. This epic builds predictive capability that gives operations teams a 15- to 60-minute advance warning, enabling proactive rather than reactive resource deployment.

- US-06 (Must Have, 13 pts): As an operations manager, I want the system to predict congestion levels for each terminal zone 15, 30, and 60 minutes ahead, so that staff and resources can be deployed before queues become unmanageable.
- US-07 (Should Have, 5 pts): As a data scientist, I want the prediction model to incorporate flight schedule data (arrivals, departures, load factors), so that forecasts account for expected passenger surges.
- US-08 (Should Have, 5 pts): As a data scientist, I want a model performance monitoring dashboard that tracks prediction accuracy and detects drift, so that retraining can be triggered before accuracy degrades significantly.
- US-09 (Could Have, 8 pts): As a data scientist, I want an automated retraining pipeline that incorporates new historical data on a scheduled basis, so that the model adapts to changing passenger patterns without manual intervention.

EP-03: Queue Management Optimization

Epic Goal: Translate density data and predictions into actionable queue management recommendations.

Raw density numbers and predictions only create value when they inform concrete operational decisions — which lanes to open, where to add staff, when to activate overflow queues. This epic bridges analytics and action.

- US-10 (Must Have, 8 pts): As a terminal supervisor, I want the system to recommend which queue lanes to open or close based on current and predicted density, so that passenger throughput is maximized without over-

staffing.

- US-11 (Should Have, 5 pts): As a security checkpoint manager, I want lane-specific throughput metrics (passengers per minute) displayed alongside recommendations, so that I can validate the system's suggestions against real performance.
- US-12 (Should Have, 5 pts): As an operations manager, I want the system to recommend staff redeployment across zones when imbalances are detected, so that idle capacity in one area can relieve pressure in another.
- US-13 (Could Have, 3 pts): As a terminal supervisor, I want to provide feedback on whether a recommendation was accepted or rejected, so that the system can learn from operational decisions over time.

EP-04: Dynamic Resource Allocation

Epic Goal: Enable data-driven allocation of staff, equipment, and gates based on predicted passenger loads.

Beyond queue-lane decisions, airports must allocate broader resources — staffing shifts, equipment positioning, gate assignments. This epic extends optimization from the queue level to the operational planning level.

- US-14 (Should Have, 8 pts): As an operations manager, I want shift-staffing recommendations based on predicted passenger volumes for the next 4-8 hours, so that labor scheduling aligns with anticipated demand.
- US-15 (Should Have, 5 pts): As an airline operations coordinator, I want boarding-gate congestion predictions shared via API, so that boarding sequences can be adjusted to prevent jetway crowding.
- US-16 (Could Have, 5 pts): As a security checkpoint manager, I want equipment utilization analytics (X-ray machines, body scanners), so that maintenance windows can be scheduled during predicted low-traffic periods.
- US-17 (Could Have, 3 pts): As an operations manager, I want a what-if scenario tool that simulates congestion impact of closing a checkpoint lane, so that planned maintenance can be scheduled with minimal passenger impact.

EP-05: Real-Time Alerting & Escalation

Epic Goal: Automatically notify relevant stakeholders when congestion thresholds are breached or predicted to breach.

Even the best predictions are worthless if they never reach the people who can act. This epic ensures alerts flow through the channels operations teams actually monitor, with escalation logic that prevents critical situations from being overlooked.

- US-18 (Must Have, 5 pts): As a terminal supervisor, I want to receive real-time alerts on the dashboard when a zone's passenger density exceeds a configurable threshold, so that I can respond immediately.
- US-19 (Should Have, 5 pts): As a terminal supervisor, I want alerts delivered

via mobile push notification and SMS, so that I am notified even when away from the dashboard.

- US-20 (Should Have, 3 pts): As an operations manager, I want unacknowledged critical alerts to escalate automatically to senior staff after a configurable time window, so that urgent situations are never missed.
- US-21 (Should Have, 3 pts): As a system administrator, I want alert delivery status (sent, delivered, acknowledged) logged for every notification, so that communication failures can be diagnosed.

EP-06: Passenger Flow Analytics & Reporting

Epic Goal: Provide historical analytics and executive dashboards to support strategic planning and performance tracking.

Real-time operations are only half the picture. Airport leadership needs trend analysis, peak-period identification, and service-level tracking to inform infrastructure investments, staffing contracts, and passenger experience initiatives.

- US-22 (Should Have, 5 pts): As an operations manager, I want historical passenger flow reports segmented by zone, day of week, and time of day, so that I can identify recurring congestion patterns.
- US-23 (Should Have, 5 pts): As an airport executive, I want a KPI dashboard showing average wait times, queue lengths, and service-level attainment, so that operational performance can be tracked against targets.
- US-24 (Could Have, 3 pts): As an operations manager, I want to compare current-day flow against the same day in prior weeks, so that anomalies can be identified quickly.
- US-25 (Could Have, 3 pts): As a passenger experience officer, I want dwell-time analytics for retail and lounge areas, so that concession partners can optimize staffing and promotions.
- US-26 (Could Have, 3 pts): As an airport executive, I want an automated weekly summary report emailed as a PDF, so that leadership stays informed without logging into the dashboard daily.

EP-07: Security Checkpoint Optimization

Epic Goal: Improve security screening throughput while maintaining compliance with safety and regulatory protocols.

Security checkpoints are the most common bottleneck in passenger flow. This epic applies the platform's monitoring and prediction capabilities specifically to security operations, where even small throughput improvements yield outsized benefits for overall passenger experience.

- US-27 (Should Have, 8 pts): As a security checkpoint manager, I want a dedicated checkpoint dashboard showing lane-by-lane throughput, queue depth, and wait-time estimates, so that I can manage capacity in real time.
- US-28 (Should Have, 5 pts): As a security checkpoint manager, I want

predictive alerts specifically for security checkpoint congestion, so that lanes can be opened proactively before queues extend into the terminal.

- US-29 (Could Have, 5 pts): As a security checkpoint manager, I want diversion recommendations suggesting passengers be routed to alternate checkpoints when one is overloaded, so that total screening capacity is balanced.
- US-30 (Could Have, 3 pts): As a security checkpoint manager, I want screening-time analytics by lane and time period, so that slower lanes or equipment issues can be identified for improvement.

EP-08: System Security, Access Control & Privacy Compliance

Epic Goal: Protect passenger data, enforce role-based access, and ensure the platform meets privacy regulations.

A platform processing video feeds across an airport handles sensitive data at scale. This epic ensures that access is tightly controlled, actions are auditable, and privacy-preserving design principles are enforced from the start.

- US-31 (Must Have, 5 pts): As a system administrator, I want role-based access control for different user types (operations, security, IT, executive), so that users only access features and data permitted for their role.
- US-32 (Must Have, 3 pts): As a compliance officer, I want a complete audit log of all data access, configuration changes, and alert acknowledgments, so that the platform can demonstrate compliance with privacy and operational regulations.
- US-33 (Must Have, 5 pts): As a privacy officer, I want the computer vision pipeline to process video using anonymized silhouette detection with no facial feature extraction, so that passenger identities are never captured or stored.
- US-34 (Should Have, 3 pts): As a system administrator, I want data retention policies enforced automatically (e.g., raw video purged after 72 hours, aggregated analytics retained for 2 years), so that storage costs and privacy exposure are minimized.

3. Acceptance Criteria (Given-When-Then)

Every user story has clear, testable acceptance criteria written in the Given-When-Then format. For Must-Have stories, a second edge-case scenario is included to verify robustness against failure conditions.

Story	Given	When	Then
US-01 (Scenario 1)	a terminal camera is online and streaming video	the ingestion service connects to the camera feed	video frames are received, decoded, and queued for processing within 2 seconds of capture.

US-01 (Scenario 2)	a terminal camera becomes unreachable (network failure)	the ingestion service detects the connection loss	the system logs the failure, raises an infrastructure alert, and continues processing feeds from other cameras without crashing.
US-02 (Scenario 1)	an IoT infrared counter at a queue entry point registers a passenger passing	the sensor publishes a count event to the ingestion pipeline	the event is recorded with timestamp and zone ID within 1 second and is available for cross-referencing.
US-02 (Scenario 2)	an IoT sensor fails to publish any events for 5 minutes (sensor timeout)	the monitoring service evaluates sensor health	the system logs the timeout, raises a sensor-health alert, and marks that sensor's data as stale in downstream calculations.
US-03 (Scenario 1)	a video frame from a terminal zone is received by the computer vision pipeline	the pipeline processes the frame	a passenger density estimate (count per defined area) is produced using anonymized silhouette detection, with no facial features extracted or stored.
US-03 (Scenario 2)	a video frame is received with severe lighting issues (overexposure or near-darkness)	the pipeline attempts density estimation	the system flags the frame as low-confidence, publishes a degraded-quality warning, and uses the last reliable estimate until quality improves.
US-04	zone-level density data is being processed continuously	30 seconds have elapsed since the last zone-count update	a new zone-count record is published to the real-time dashboard, timestamped within the current 30-second window.
US-05	the ingestion pipeline encounters a dropped video frame or sensor timeout	the issue is detected during processing	the system logs the issue with timestamp, zone, and severity, and the log is accessible via the admin monitoring interface.
US-06 (Scenario 1)	sufficient historical and real-time density data exists for a terminal zone	the prediction engine runs its scheduled forecast job	the system outputs congestion predictions for 15-, 30-, and 60-minute horizons with confidence

			intervals, refreshed at least every 15 minutes.
US-06 (Scenario 2)	a zone has just been added to the system with no historical data	the prediction engine attempts to forecast	the system withholds a numeric prediction for that zone, displays an "insufficient data" status, and relies on rule-based heuristics until sufficient history accumulates.
US-07	flight schedule data (arrivals, departures, load factors) is available from FIDS integration	the prediction model generates a forecast	the model incorporates scheduled flight data as a feature, improving forecast accuracy for expected passenger surges.
US-08	the production prediction model has been running for a configured evaluation period	the monitoring service compares live predictions against observed outcomes	the system calculates prediction accuracy metrics and flags the model as "drifted" if accuracy falls below a configured threshold.
US-09	new historical data is available and the scheduled retraining window has arrived	the retraining pipeline executes	the model is retrained, validated against a holdout dataset, and only promoted to production if accuracy meets or exceeds the current model.
US-10 (Scenario 1)	current and predicted density data is available for a security checkpoint zone	the queue optimization engine evaluates the data	the system generates a lane-opening/closing recommendation with supporting rationale, displayed on the terminal supervisor dashboard.
US-10 (Scenario 2)	all checkpoint lanes are already open and congestion remains above threshold	the optimization engine evaluates options	the system indicates no additional lanes can be opened, recommends staff redeployment from other zones, and raises a capacity-limit warning.
US-11	lane-specific throughput data (passengers per	a terminal supervisor views the queue	throughput metrics are displayed per lane alongside active recommendations,

	minute) is being collected	management dashboard	updated in real time.
US-12	passenger density is significantly higher in Zone A than Zone B, and Zone B has idle staff	the resource optimization engine evaluates cross-zone capacity	the system recommends redeploying N staff from Zone B to Zone A, with estimated impact on wait times.
US-13	a recommendation has been displayed to a terminal supervisor	the supervisor marks the recommendation as "accepted" or "rejected"	the feedback is logged with timestamp and user ID, and is available for model/recommendation tuning.
US-14	predicted passenger volumes for the next 4-8 hours are available	a shift-scheduling job runs	the system generates staffing recommendations per zone and shift, aligned with predicted demand.
US-15	boarding-gate congestion predictions are generated for an upcoming departure	an airline operations system queries the API	the API returns the predicted congestion level and recommended boarding-sequence adjustments.
US-16	equipment utilization data (active/idle time) is collected for security screening equipment	a security checkpoint manager views the equipment analytics screen	utilization percentages are displayed per device, and low-traffic windows are highlighted for potential maintenance scheduling.
US-17	the operations manager wants to simulate closing one checkpoint lane	they select a lane and run a what-if scenario	the system estimates the congestion impact (additional wait time, queue length increase) and displays the results before any real action is taken.
US-18 (Scenario 1)	a zone's passenger density exceeds the configured alert threshold	the alerting engine evaluates the threshold rule	an alert is generated and displayed on the terminal supervisor dashboard within 30 seconds, with severity level and recommended actions.
US-18 (Scenario 2)	density briefly exceeds the	the alerting engine evaluates	the alert is suppressed or auto-resolved, and an entry is

io 2)	threshold due to a sensor data spike that is corrected in the next reading	the corrected data	logged for audit purposes.
US-20	a critical alert has been sent to a terminal supervisor and remains unacknowledged after the configured time (e.g., 10 minutes)	the escalation checker runs	the alert is automatically escalated to the operations manager and marked as "escalated" in the alert history.
US-21	an alert notification is sent via any channel (dashboard, push, SMS)	delivery is attempted	the system logs delivery status (sent, delivered, failed, acknowledged) for each channel and recipient.
US-22	historical density and flow data exists for at least 30 days	an operations manager requests a historical flow report	the system generates a report segmented by zone, day of week, and time of day, available for download or in-app viewing.
US-23	real-time and historical KPI data is available	an airport executive opens the KPI dashboard	a summarized view of average wait times, queue lengths, and service-level attainment is displayed, refreshed in real time.
US-24	current-day flow data is available, along with data from the same day in prior weeks	an operations manager views the comparison screen	the system displays a side-by-side comparison chart highlighting deviations from historical norms.
US-25	dwell-time data has been collected for retail and lounge zones	a passenger experience officer opens the dwell-time analytics screen	average and distribution of dwell times are displayed per zone, segmented by time of day.
US-26	a weekly summary report is scheduled and the reporting job runs	the scheduled time is reached	a PDF summarizing key KPIs, congestion events, and trends from the prior week is generated and emailed to subscribed executives.

US-27	lane-level throughput, queue depth, and wait-time data is available for a security checkpoint	a security checkpoint manager opens the checkpoint dashboard	a dedicated view displays per-lane metrics, updated in real time, with visual indicators for performance thresholds.
US-28	congestion is predicted to exceed threshold at a security checkpoint within 15 minutes	the alerting engine evaluates checkpoint predictions	a predictive alert is generated and displayed on the checkpoint manager's dashboard, with recommended preemptive actions.
US-29	one security checkpoint is overloaded while another has available capacity	the diversion engine evaluates checkpoint loads	the system recommends routing a portion of passengers to the less-congested checkpoint, with estimated impact on both queues.
US-30	screening-time data is collected per lane and time period	a security checkpoint manager views the screening analytics screen	average screening times are displayed per lane, with outliers and trends highlighted.
US-31 (Scenario 1)	a new user account is created for the platform	an administrator assigns a role (e.g., terminal supervisor, executive)	the system restricts that user's access strictly to the features and data permitted for their assigned role.
US-31 (Scenario 2)	a user attempts to access a feature outside their assigned role's permissions	the access-control middleware evaluates the request	the system denies access, returns a clear permission-denied message, and logs the attempted access in the audit trail.
US-32 (Scenario 1)	any user views, creates, or modifies a record, or acknowledges an alert	the action is performed	the system logs the user ID, timestamp, action type, and affected record in an immutable audit trail.
US-32 (Scenario 2)	an attempt is made to directly modify or delete an existing audit-log entry	the modification request reaches the audit-log storage layer	the system rejects the modification (audit log is append-only) and raises a security alert to the

US-33	a video frame is processed by the computer vision pipeline	the pipeline extracts density data	compliance officer. only anonymized silhouette/ blob detection is used; no facial features are extracted, and no identifiable imagery is stored beyond the raw video retention window.
US-34	the data retention window for raw video (e.g., 72 hours) has elapsed	the retention enforcement job runs	raw video older than the retention window is purged automatically, and a log entry confirms the deletion.

4. Product Backlog Prioritization

4.1 Prioritized Product Backlog (MoSCoW)

The backlog was prioritized using MoSCoW, informed by business value, dependency order, and regulatory urgency. Foundational data-ingestion, privacy-preserving vision, and security stories are ranked highest since every other capability depends on them.

Rank	Story	Epic	MoSCoW	Business Value	Dependency
1	US-01	Real-Time Passenger Monitoring	Must Have	High	None (foundational)
2	US-02	Real-Time Passenger Monitoring	Must Have	High	None (foundational)
3	US-03	Real-Time Passenger Monitoring	Must Have	High	US-01
4	US-33	System Security & Privacy	Must Have	High	US-03
5	US-31	System Security & Privacy	Must Have	High	None (foundational)
6	US-06	AI-Based Congestion Prediction	Must Have	High	US-01, US-02, US-03
7	US-10	Queue Management Optimization	Must Have	High	US-06
8	US-18	Real-Time Alerting	Must Have	High	US-06

9	US-32	System Security & Privacy	Must Have	High	US-31
11	US-07	AI-Based Congestion Prediction	Should Have	Medium	US-06, FIDS integration
12	US-08	AI-Based Congestion Prediction	Should Have	Medium	US-06
13	US-11	Queue Management Optimization	Should Have	Medium	US-10
14	US-12	Queue Management Optimization	Should Have	Medium	US-06
15	US-14	Dynamic Resource Allocation	Should Have	Medium	US-06
16	US-15	Dynamic Resource Allocation	Should Have	Medium	US-06
17	US-19	Real-Time Alerting	Should Have	Medium	US-18
18	US-20	Real-Time Alerting	Should Have	Medium	US-18
19	US-21	Real-Time Alerting	Should Have	Medium	US-18
20	US-22	Flow Analytics & Reporting	Should Have	Medium	US-01, US-02, US-03
21	US-23	Flow Analytics & Reporting	Should Have	Medium	US-22
22	US-27	Security Checkpoint Optimization	Should Have	Medium	US-03, US-06
23	US-28	Security Checkpoint Optimization	Should Have	Medium	US-06, US-27
24	US-34	System Security & Privacy	Should Have	Medium	US-01, US-33
25	US-05	Real-Time Passenger Monitoring	Should Have	Low	US-01, US-02

26	US-09	AI-Based Congestion Prediction	Could Have	Medium	US-06, US-08
27	US-13	Queue Management Optimization	Could Have	Low	US-10
28	US-16	Dynamic Resource Allocation	Could Have	Low	US-27
29	US-17	Dynamic Resource Allocation	Could Have	Low	US-06
30	US-24	Flow Analytics & Reporting	Could Have	Low	US-22
31	US-25	Flow Analytics & Reporting	Could Have	Low	US-03
32	US-26	Flow Analytics & Reporting	Could Have	Low	US-23
33	US-29	Security Checkpoint Optimization	Could Have	Low	US-27, US-28
34	US-30	Security Checkpoint Optimization	Could Have	Low	US-27

4.2 Team Capacity & Velocity Planning

Sprint sizing is based on velocity data from prior comparable projects and the team composition. A 2-week sprint cadence is adopted with a 7-person cross-functional team (2 backend/infrastructure engineers, 1 data scientist/ML engineer, 1 computer vision engineer, 1 frontend/dashboard developer, 1 QA engineer, and 1 Scrum Master/Product Owner pairing).

Reference Sprint	Velocity (Story Points)	Notes
Prior project - Sprint A	34 points	Baseline reference
Prior project - Sprint B	40 points	Baseline reference
Prior project - Sprint C	37 points	Baseline reference
Average Velocity	~37 points/sprint	Used to size Sprint 1 and Sprint 2

5. Sprint Planning

Based on the prioritized backlog, dependency chains, and team capacity, two sprints are planned. Sprint 1 establishes the foundational data pipeline, privacy-preserving computer vision, initial prediction capability, and basic alerting. Sprint 2 builds on this foundation by adding queue optimization recommendations, multi-channel alerting, security controls, and analytics dashboards.

5.1 Sprint 1 Backlog & Development Tasks

US-01: Video feed ingestion from terminal cameras (8 pts)

- Design camera feed ingestion architecture (RTSP/RTMP handling)
- Build video stream connector service with connection pooling
- Implement frame extraction and queuing (message broker integration)
- Add connection health monitoring and auto-reconnect logic
- Write unit and integration tests for ingestion pipeline

US-02: IoT sensor data ingestion (5 pts)

- Define sensor event schema (infrared counters, beacon events)
- Build sensor data ingestion API (MQTT/HTTP endpoints)
- Implement event validation and timestamping
- Add sensor health monitoring (timeout detection)
- Write unit and integration tests

US-03: Computer vision density estimation pipeline (8 pts)

- Set up computer vision model serving infrastructure (GPU-enabled)
- Implement anonymized silhouette detection model (no facial features)
- Build frame-processing service with density output
- Implement confidence scoring for low-quality frames
- Write tests with sample video data

US-33: Privacy-preserving design enforcement (5 pts)

- Implement facial-feature blocking safeguard in vision pipeline
- Add validation tests confirming no facial data extraction
- Document privacy controls for compliance review
- Conduct privacy impact assessment review

US-31: Role-based access control (5 pts)

- Design role and permission model (operations, security, IT, executive)
- Implement authentication and authorization middleware

- Build admin UI for user and role management
- Restrict API endpoints and UI components by role
- Write access-control tests

US-06: AI congestion prediction engine (13 pts) (started, continued in Sprint 2)

- Prepare and clean training dataset (historical density + FIDS)
- Select baseline time-series prediction model architecture
- Begin model training and initial validation
- Design prediction API schema for dashboard/alerting consumption

Total committed story points: 44 (Note: US-06 is partially complete; 8 points committed for Sprint 1, with 5 points carrying into Sprint 2)

Adjusted Sprint 1 commitment: 39 points (accounting for US-06 split)

5.2 Sprint 2 Backlog & Development Tasks

US-06: AI congestion prediction engine (continued, 5 pts)

- Complete model training and validation
- Build scheduled inference job
- Expose prediction API for dashboard and alerting consumption
- Write prediction accuracy tests

US-10: Queue lane optimization recommendations (8 pts)

- Define recommendation rules engine (lane open/close logic)
- Build recommendation generation service consuming prediction API
- Design recommendation display component for supervisor dashboard
- Implement recommendation rationale logging
- Write tests for edge cases (all lanes open, capacity limit)

US-18: Real-time dashboard alerting (5 pts)

- Define configurable threshold rules per zone
- Build alert generation service triggered by density/predictions
- Build in-dashboard alert notification component
- Log alert history with timestamps and severity
- Write threshold edge-case tests

US-32: Audit logging (3 pts)

- Design immutable audit-log schema
- Implement logging hooks on all data access, modifications, and alert

acknowledgments

- Build audit-log viewer for compliance officers
- Add tamper-evidence (append-only storage, hashing)
- Write compliance-focused test cases

US-04: Zone-level counts updated every 30 seconds (5 pts)

- Implement aggregation job for zone-level counts
- Ensure dashboard receives updates within 30-second windows
- Add timestamp validation in data pipeline
- Write latency and accuracy tests

US-07: Flight schedule integration for predictions (5 pts)

- Integrate with FIDS API for arrival/departure data
- Transform flight data into prediction model features
- Update model training pipeline to incorporate flight data
- Write integration tests with mock FIDS data

US-19: Multi-channel alerting (mobile push, SMS) (5 pts)

- Integrate mobile push notification service
- Integrate SMS gateway provider
- Build multi-channel dispatch service
- Implement delivery-status logging
- Test failover between channels

Total committed story points: 36

5.3 Future Backlog (Sprint 3+ Candidates)

The remaining items extend model adaptability, resource allocation depth, analytics richness, and security checkpoint specialization. These are scheduled for future sprints once the core monitoring-prediction-alerting loop and compliance foundation are validated.

- US-08 – Model performance monitoring dashboard (5 pts)
- US-09 – Automated model retraining pipeline (8 pts)
- US-11 – Lane-specific throughput metrics (5 pts)
- US-12 – Cross-zone staff redeployment recommendations (5 pts)
- US-13 – Recommendation feedback logging (3 pts)
- US-14 – Shift-staffing recommendations (8 pts)
- US-15 – Boarding-gate congestion API for airlines (5 pts)

- US-16 – Equipment utilization analytics (5 pts)
- US-17 – What-if scenario simulation (3 pts)
- US-20 – Alert escalation automation (3 pts)
- US-21 – Alert delivery status logging (3 pts)
- US-22 – Historical flow reports (5 pts)
- US-23 – Executive KPI dashboard (5 pts)
- US-24 – Same-day historical comparison (3 pts)
- US-25 – Dwell-time analytics for retail zones (3 pts)
- US-26 – Automated weekly PDF report (3 pts)
- US-27 – Security checkpoint dedicated dashboard (8 pts)
- US-28 – Checkpoint-specific predictive alerts (5 pts)
- US-29 – Checkpoint diversion recommendations (5 pts)
- US-30 – Screening-time analytics (3 pts)
- US-34 – Data retention policy enforcement (3 pts)

6. Story Point Estimation (Fibonacci / Planning Poker)

The team used Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13) to estimate relative effort and complexity. Divergent estimates triggered discussion before consensus.

Story	Epic	Points	Justification
US-01	Real-Time Passenger Monitoring	8	High complexity: real-time video streaming infrastructure, multi-source ingestion, connection resilience.
US-02	Real-Time Passenger Monitoring	5	Moderate complexity: IoT protocol handling, sensor health monitoring.
US-03	Real-Time Passenger Monitoring	8	High complexity: computer vision model deployment, GPU infrastructure, anonymization enforcement.
US-04	Real-Time Passenger Monitoring	5	Moderate complexity: aggregation logic, latency optimization.

US-05	Real-Time Passenger Monitoring	3	Small scope: logging and monitoring enhancements.
US-06	AI-Based Congestion Prediction	13	Very high complexity: data preparation, model selection, training, validation, inference pipeline.
US-07	AI-Based Congestion Prediction	5	Moderate complexity: API integration, feature engineering.
US-08	AI-Based Congestion Prediction	5	Moderate complexity: metrics pipeline, drift detection.
US-09	AI-Based Congestion Prediction	8	High complexity: automated retraining, validation, promotion pipeline.
US-10	Queue Management Optimization	8	High complexity: rules engine, multi-factor recommendation logic.
US-11	Queue Management Optimization	5	Moderate complexity: metrics aggregation, dashboard integration.
US-12	Queue Management Optimization	5	Moderate complexity: cross-zone logic, staffing recommendations.
US-13	Queue Management Optimization	3	Small scope: feedback capture, logging.
US-14	Dynamic Resource Allocation	8	High complexity: shift-planning logic, demand forecasting integration.
US-15	Dynamic Resource Allocation	5	Moderate complexity: API exposure, airline integration contract.
US-16	Dynamic Resource Allocation	5	Moderate complexity: equipment telemetry, utilization calculation.

US-17	Dynamic Resource Allocation	3	Small scope: simulation UI, impact estimation.
US-18	Real-Time Alerting	5	Moderate complexity: threshold engine, dashboard notification.
US-19	Real-Time Alerting	5	Moderate complexity: multi-channel integration, delivery logging.
US-20	Real-Time Alerting	3	Small scope: escalation timer, notification routing.
US-21	Real-Time Alerting	3	Small scope: delivery status logging, monitoring.
US-22	Flow Analytics & Reporting	5	Moderate complexity: aggregation queries, report generation.
US-23	Flow Analytics & Reporting	5	Moderate complexity: dashboard design, KPI calculation.
US-24	Flow Analytics & Reporting	3	Small scope: comparison logic, charting.
US-25	Flow Analytics & Reporting	3	Small scope: dwell-time calculation, zone segmentation.
US-26	Flow Analytics & Reporting	3	Small scope: PDF generation, email scheduling.
US-27	Security Checkpoint Optimization	8	High complexity: lane-level data, dedicated UI, real-time updates.
US-28	Security Checkpoint Optimization	5	Moderate complexity: checkpoint-specific prediction integration.
US-29	Security Checkpoint Optimization	5	Moderate complexity: diversion logic, multi-checkpoint balancing.
US-30	Security Checkpoint Optimization	3	Small scope: screening-time aggregation, analytics display.

US-31	System Security & Privacy	5	Moderate complexity: RBAC middleware, admin UI.
US-32	System Security & Privacy	3	Small scope: audit logging hooks, viewer UI.
US-33	System Security & Privacy	5	Moderate complexity: privacy enforcement in vision pipeline, validation.
US-34	System Security & Privacy	3	Small scope: retention job, automated purging.

Sprint 1 committed effort: 39 story points. Sprint 2 committed effort: 36 story points.
Full backlog total estimated effort: 175 story points across 34 user stories.

7. Sprint Goals and Expected Deliverables

7.1 Sprint 1 Goal

By the end of Sprint 1, deliver a working data pipeline that ingests real-time video and IoT sensor data, processes frames through a privacy-preserving computer vision model to estimate passenger density, establishes role-based access control, and begins generating AI-based congestion predictions — laying the foundation for the airport's intelligent flow management platform.

Sprint 1 Expected Deliverables

- Functional video feed ingestion service connected to terminal cameras.
- Functional IoT sensor data ingestion API receiving queue counter events.
- Deployed computer vision pipeline producing anonymized density estimates per zone.
- Privacy enforcement validated (no facial feature extraction or storage).
- Role-based access control enforced across platform features and APIs.
- Initial congestion prediction model trained and prediction API schema defined.
- Sprint demo showing: video ingestion density estimation initial prediction output.
- Updated product backlog and burndown chart at Sprint Review.

7.2 Sprint 2 Goal

By the end of Sprint 2, extend the platform with zone-level counts updated every 30 seconds, a fully deployed AI prediction engine with flight-schedule integration, queue lane optimization recommendations, real-time dashboard alerting with multi-channel delivery, and a compliance-ready audit trail — advancing the system from a data-ingestion prototype toward an operationally actionable congestion management tool.

Sprint 2 Expected Deliverables

- Zone-level passenger counts updated and displayed within 30-second intervals.
- Fully deployed prediction model generating 15/30/60-minute forecasts.
- Flight schedule data integrated into prediction pipeline.
- Queue lane optimization recommendations displayed on supervisor dashboard.
- Real-time threshold-based alerts visible on dashboard.
- Multi-channel alert delivery (mobile push + SMS) with logged delivery status.
- Audit log covering all data access, modifications, and alert acknowledgments.
- Updated burndown chart, backlog, and compliance checklist at Sprint Review.

8. Definition of Ready and Definition of Done

8.1 Definition of Ready (DoR)

A backlog item is considered ready to be pulled into a sprint only when:

- The user story follows the "As a / I want / so that" format.
- Acceptance criteria are written in Given-When-Then format and agreed upon by the Product Owner.
- Dependencies on other stories, external systems (FIDS, camera network, IoT sensors), or infrastructure are identified and either resolved or explicitly scheduled.
- The story has been estimated by the team using Planning Poker.
- Required design assets, API contracts, data schemas, or infrastructure access are available or scoped.
- The story is small enough to be completed within a single sprint.
- Privacy and compliance implications have been reviewed.

8.2 Definition of Done (DoD)

A user story is considered complete only when:

- Code is peer-reviewed, merged into the main branch, and passes all automated tests.
- Acceptance criteria are independently verified by QA.
- The feature is deployed to the staging environment and demonstrated at Sprint Review.
- Relevant documentation (API docs, user guides, data schemas, privacy controls) is updated.

- No known critical or high-severity defects remain open against the story.
- Security and access-control implications have been reviewed and addressed.
- Privacy compliance has been verified for any story involving video or sensor data.
- The Product Owner has formally accepted the story.

9. Risk Management and Mitigation Plan

Risk	Likelihood	Impact	Mitigation Strategy
Camera network instability or coverage gaps	High	High	Implement auto-reconnect logic with fallback to IoT sensor data; prioritize high-traffic zones for camera coverage; monitor connection health with alerting.
Computer vision model produces inaccurate density estimates	Medium	High	Validate model against ground-truth counts during pilot phase; implement confidence scoring; flag low-quality frames; retrain with airport-specific data.
Privacy concerns over video-based monitoring	Medium	High	Enforce anonymized silhouette detection from day one; conduct privacy impact assessment; implement strict data retention policies; engage compliance officer throughout.
IoT sensor failures or data gaps	Medium	Medium	Cross-validate sensor data with camera estimates; implement sensor health monitoring (US-05); maintain manual count fallback procedures.
Flight schedule data unavailable or delayed from FIDS	Medium	Medium	Design prediction model to function (with reduced accuracy) without flight data; implement FIDS connection health monitoring; cache last-known schedule.
Alert fatigue from excessive notifications	Medium	Medium	Tune threshold sensitivity with operations input; implement severity levels; use escalation only for unacknowledged critical alerts; allow user threshold customization.
Integration complexity with legacy airport systems	Medium	Medium	Adopt standard APIs (REST, MQTT); run integration spikes early; provide manual data-entry fallback; document integration contracts.

Team unfamiliarity with computer vision tooling	Low	Medium	Allocate time for technical spike on CV library; pair less-experienced developers with CV engineer; leverage pre-trained models where possible.
Scope creep from stakeholder requests mid-sprint	Medium	Medium	Enforce sprint backlog stability during sprint; route new requests to Product Owner for backlog grooming; maintain strict prioritization discipline.
Data retention policy non-compliance	Low	High	Implement automated retention enforcement (US-34) with audit logging; conduct compliance review before production deployment.

10. Agile Ceremonies Schedule

The team follows a standard two-week Scrum cadence.

Ceremony	Frequency	Duration	Participants	Purpose
Sprint Planning	Start of each sprint	2-4 hours	Full Scrum team, Product Owner	Select backlog items, define sprint goal, break stories into tasks.
Daily Stand-up	Daily	15 minutes	Development team, Scrum Master	Sync on progress, plans, and blockers.
Backlog Refinement	Mid-sprint, weekly	1 hour	Product Owner, development leads	Clarify, estimate, and re-prioritize upcoming backlog items.
Sprint Review	End of each sprint	1-2 hours	Full team, Product Owner, key stakeholders	Demo completed increment; gather stakeholder feedback.
Sprint Retrospective	End of each sprint (after review)	1 hour	Development team, Scrum Master	Reflect on process; identify improvements for next sprint.

11. Sprint Review and Retrospective Framework

11.1 Sprint Review Format

At the end of each sprint, the team demonstrates the working increment to stakeholders, focusing on stories that meet the Definition of Done. Feedback is logged as new or updated backlog items rather than implemented ad hoc.

Review Agenda:

1. Sprint goal recap and summary of completed stories.
2. Live demonstration of new features (video ingestion, density visualization, predictions, alerts, dashboards).
3. Stakeholder Q&A and feedback collection.
4. Backlog impact discussion (new items, priority changes).
5. Next sprint preview.

11.2 Sprint Retrospective Format

Immediately following the review, the development team holds a retrospective using the "What went well / What didn't go well / Action items" format:

- What went well: Practices, tools, or collaboration patterns to continue.
- What didn't go well: Blockers, process gaps, or friction points encountered.
- Action items: Concrete, owned, time-bound changes to try in the next sprint.

Action items are added to the following sprint's backlog and reviewed for completion at the next retrospective.

12. Glossary of Key Terms

Term	Definition
Epic	A large body of work broken into smaller, related user stories.
User Story	A short, plain-language feature description from the user's perspective.
Acceptance Criteria	Specific, testable conditions a story must satisfy to be complete.
Given-When-Then	Structured format describing context, action, and expected outcome.
Product Backlog	Prioritized list of all features, fixes, and work needed for the product.
Sprint Backlog	Subset of product backlog committed for the current sprint.
MoSCoW	Prioritization technique: Must Have, Should Have, Could Have, Won't Have.
Story Point	Relative measure of effort and complexity for a user story.
Planning Poker	Consensus-based estimation technique using Fibonacci cards.

Velocity	Average story points completed per sprint; used to forecast capacity.
Sprint Goal	Short statement of the sprint's overarching objective.
Definition of Ready (DoR)	Criteria a backlog item must meet before entering a sprint.
Definition of Done (DoD)	Criteria work must meet to be considered complete.
Computer Vision	Technology enabling machines to interpret visual data from cameras.
IoT Sensor	Internet-connected device (infrared counter, beacon) collecting physical-world data.
FIDS	Flight Information Display System — source of arrival/ departure schedules.
Congestion Threshold	Configured density level triggering an automated alert.
RACI Matrix	Responsibility chart: Responsible, Accountable, Consulted, Informed.
Silhouette Detection	Privacy-preserving computer vision technique identifying human shapes without facial features.
Dwell Time	Duration a passenger spends in a specific zone.
Throughput	Number of passengers processed per unit time (e.g., per minute).

13. Appendix A: Communication Plan

Communication Item	Audience	Frequency	Channel
Daily Stand-up Notes	Development team	Daily	Internal chat (async summary)
Sprint Review Demo & Feedback	All stakeholders, Product Owner	End of sprint	Video call + recorded demo
Congestion Alerts	Terminal supervisors, operations managers, checkpoint managers	Real-time (event-driven)	Dashboard + Mobile push + SMS

Weekly Executive Summary	Airport executives, operations leadership	Weekly	Automated PDF email report
Privacy/Compliance Updates	Compliance officer, project leadership	As needed / milestone-based	Email + scheduled review meetings

14. Appendix B: Sample Sprint 1 Burndown

Sprint Day	Remaining Story Points	Notes
Day 1	39	Sprint planning complete; tasks assigned.
Day 2	37	Infrastructure setup underway; camera connectivity spike.
Day 3	34	Video ingestion service scaffolding in progress.
Day 5	29	Camera ingestion nearing completion; IoT API started.
Day 7	23	Mid-sprint checkpoint; CV pipeline development active.
Day 9	16	CV model deployed to staging; privacy validation in progress.
Day 11	9	RBAC middleware complete; prediction model training ongoing.
Day 13	3	All stories in QA verification; minor bug fixes.
Day 14	0	All Sprint 1 stories meet Definition of Done; ready for review.

A steady decline indicates appropriate task granularity and timely impediment resolution.

15. Appendix C: Team Roles and Responsibilities

Role	Responsibilities	Key Skills
Product Owner	Owns vision, prioritizes backlog, accepts completed stories.	Airport operations domain knowledge; stakeholder management.
Scrum Master	Facilitates ceremonies, tracks velocity, removes impediments.	Agile facilitation; servant leadership; conflict resolution.

Backend/ Infrastructure Engineers (x2)	Build ingestion pipelines, alerting services, APIs, integration middleware.	Real-time streaming; message brokers; REST/ MQTT; cloud infrastructure.
Data Scientist / ML Engineer	Develops, validates, monitors congestion prediction models.	Time-series forecasting; ML model validation; Python/ML frameworks.
Computer Vision Engineer	Builds and optimizes the density estimation pipeline.	Computer vision libraries; GPU optimization; privacy- preserving techniques.
Frontend/Dashboard Developer	Builds operator dashboards, mobile views, analytics visualizations.	Data visualization; responsive UI; real-time websockets.
QA Engineer	Verifies acceptance criteria, writes automated tests, validates edge cases.	Test automation; exploratory testing; performance testing.
Privacy & Compliance Officer	Reviews privacy controls, audit logging, regulatory compliance.	GDPR/data protection regulations; security auditing.

16. Appendix D: Technical Architecture Overview

The Smart Airport Passenger Flow and Queue Management System follows a layered architecture designed for real-time data processing, scalability, and privacy compliance.

Data Ingestion Layer

- **Video Stream Connectors:** RTSP/RTMP connectors receive feeds from terminal cameras, extract frames, and publish to a high-throughput message broker (e.g., Kafka, AWS Kinesis).
- **IoT Sensor Gateway:** MQTT/HTTP endpoints receive events from infrared counters, Bluetooth beacons, and Wi-Fi tracking devices, normalizing data into a common event schema.
- **FIDS Integration:** REST API connector pulls flight schedule data (arrivals, departures, load factors) on a scheduled basis and caches for prediction model consumption.

Processing Layer

- **Computer Vision Service:** GPU-enabled microservice consumes video frames, applies anonymized silhouette detection models, and outputs zone-level density estimates with confidence scores.
- **Aggregation Service:** Combines camera-derived density with IoT sensor counts, resolves discrepancies, and publishes authoritative zone-level counts

every 30 seconds.

- Prediction Engine: Time-series ML models consume historical density, current counts, and flight schedule data to generate 15/30/60-minute congestion forecasts per zone.

Application Layer

- Queue Optimization Engine: Rules-based and ML-assisted service generates lane open/close recommendations and staff redeployment suggestions.
- Alerting Service: Evaluates configurable threshold rules against density and predictions, generates alerts, and dispatches via dashboard, mobile push, and SMS.
- Analytics Engine: Aggregates historical data for trend analysis, KPI calculation, and report generation.

Presentation Layer

- Operator Dashboard (Web): Real-time zone map, density visualizations, predictions, alerts, and recommendations for terminal supervisors and operations managers.
- Executive Dashboard (Web): KPI summaries, historical trends, service-level tracking for airport leadership.
- Mobile App: Push notifications and simplified zone-status views for supervisors in the field.
- API Gateway: Exposes REST APIs for airline integration (boarding-gate predictions) and third-party systems.

Infrastructure & Security Layer

- Cloud Infrastructure: Scalable compute (containerized microservices), managed databases (time-series, relational), object storage (raw video with retention policies).
- RBAC Middleware: Enforces role-based access across all APIs and UI components.
- Audit Logging: Append-only log captures all data access, modifications, and alert acknowledgments.
- Data Retention Service: Automated purging of raw video per configured retention windows; aggregated analytics retained per policy.

17. Appendix E: Integration Points and Data Flow

Integration Point	Data Direction	Protocol	Purpose
Terminal Cameras	Inbound	RTSP/RTMP	Video feed ingestion for density estimation.

IoT Sensors (Infrared, Beacons)	Inbound	MQTT/HTTP	Queue entry/exit counts, occupancy validation.
Operator Dashboard	Outbound	WebSocket/ REST	Real-time density, predictions, alerts, recommendations.
Mobile App (Supervisors)	Outbound	Push Notification / REST	Alerts and zone-status summaries.
SMS Gateway	Outbound	API	Alert delivery for critical notifications.
Airline Operations API	Outbound	REST API	Boarding-gate congestion predictions for boarding- sequence adjustments.
Audit Log Storage	Internal	Append-only DB	Compliance logging of all access and modifications.

Data Flow Summary

Cameras & IoT Sensors Ingestion Layer (Kafka/Kinesis)
 Computer Vision Service Density Estimates
 IoT Event Processing

Aggregation Service (Zone Counts)

Engine Prediction Engine Queue Optimization Analytics
 Engine

Alerting Service

(Web) Mobile App SMS/Email Dashboard